

Introduction to astronomical spectroscopy

M1 CCP HAP 703P

Atelier Astrophysique 1

Julien Morin

21 October 2026

FACULTÉ DES SCIENCES DE MONTPELLIER

LUPM LABORATOIRE MONTPELLIER UNIVERS & PARTICULES

Outline

1 Spectroscopy in Astrophysics

2 Optical-range spectrometers

3 Bibliographic ressources

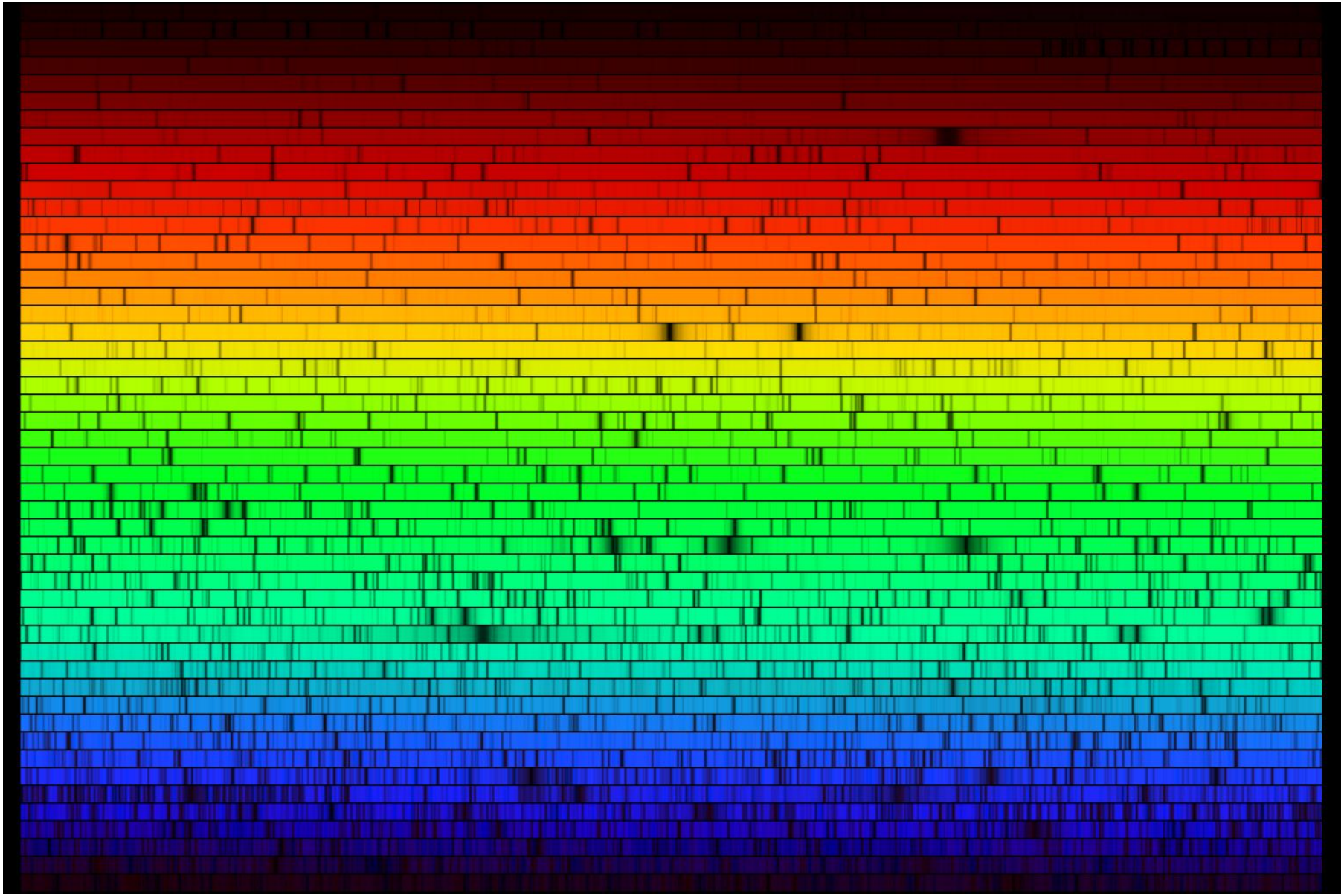
Outline

1 Spectroscopy in Astrophysics
- Stellar spectra
- Spectral type, peculiarities and abundances
- Radial velocities and redshifts
- Magnetic field measurements
- Importance of spectral resolution

2 Optical-range spectrometers

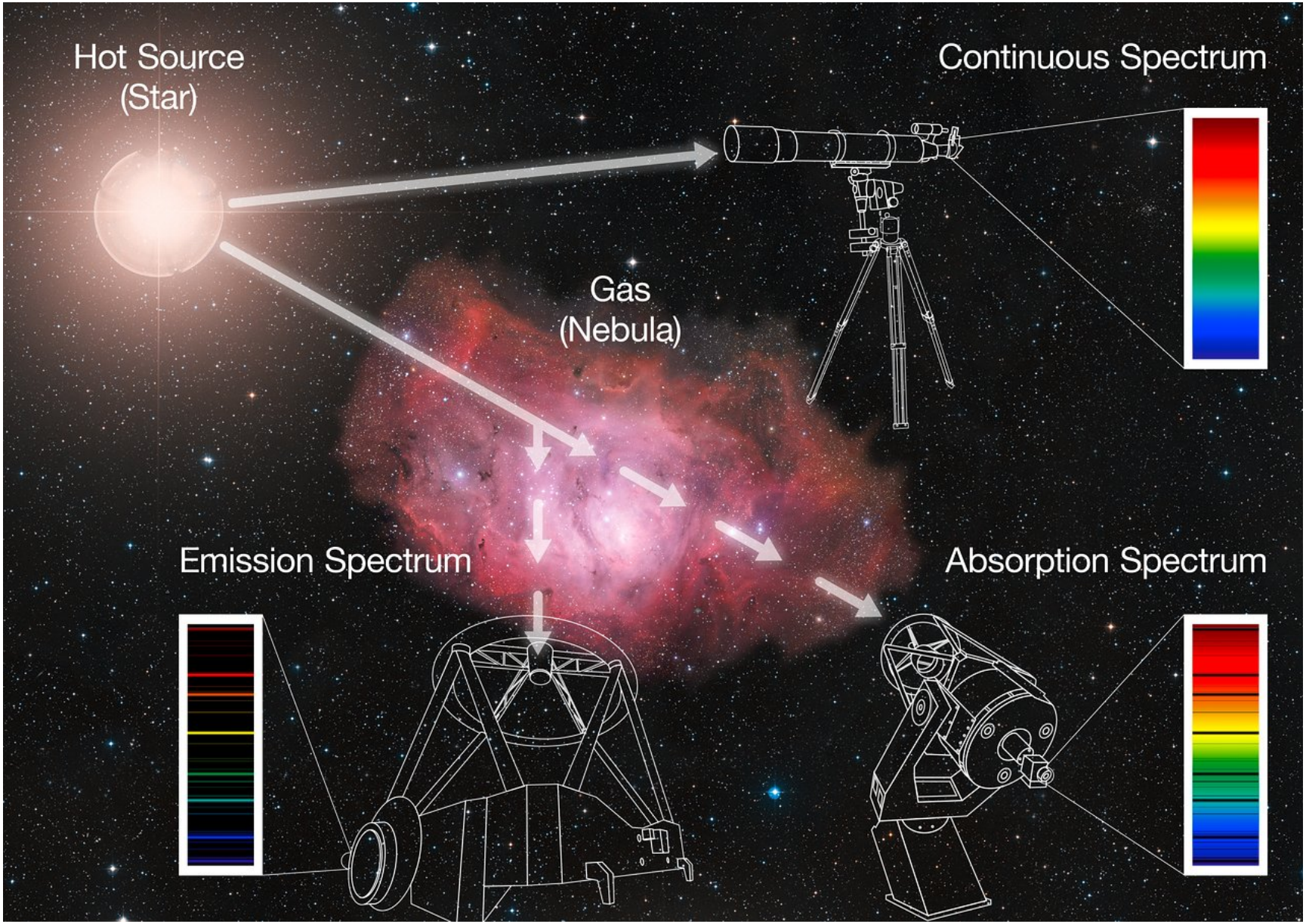
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Stellar spectra



Solar spectrum, credit: NOAO

Stellar spectra

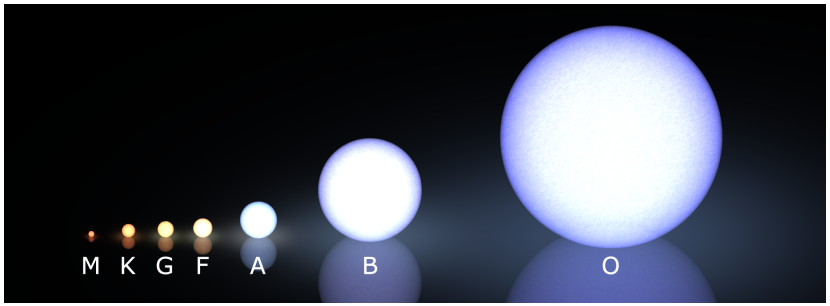


Formation of spectra: Kirchhoff's laws, credit: ESO

Spectral type, peculiarities and abundances

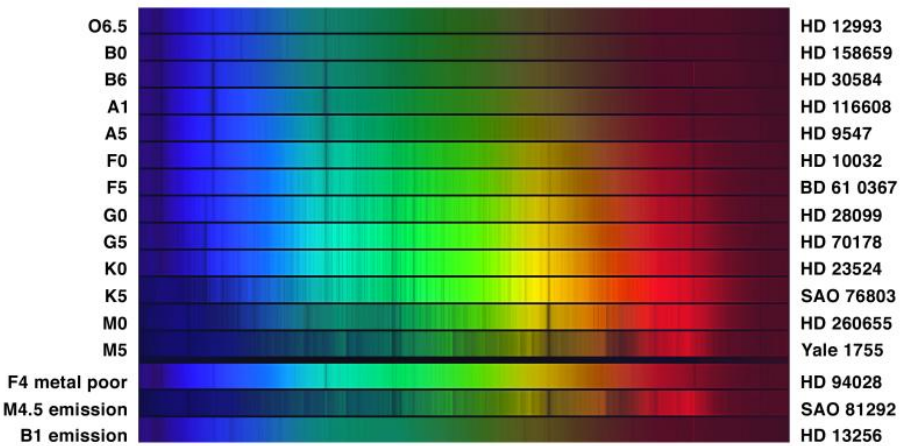
- $T_{\rm eff}$, $\log g$, [Fe/H], [X/H]
  - Note:
    $$[\mathrm{X/H}] = \log_{10}\left(\frac{n_{\rm Fe}}{n_{\rm H}}\right) - \log_{10}\left(\frac{n_{\rm Fe}}{n_{\rm H}}\right)_{\odot}$$
- SpT OBAFGKM(RNS/LTY)
- Luminosity class I➔V
- Peculiarities:
  - composition
  - emission lines

➔ p, m, e, ...



Artist impression of the MK classification

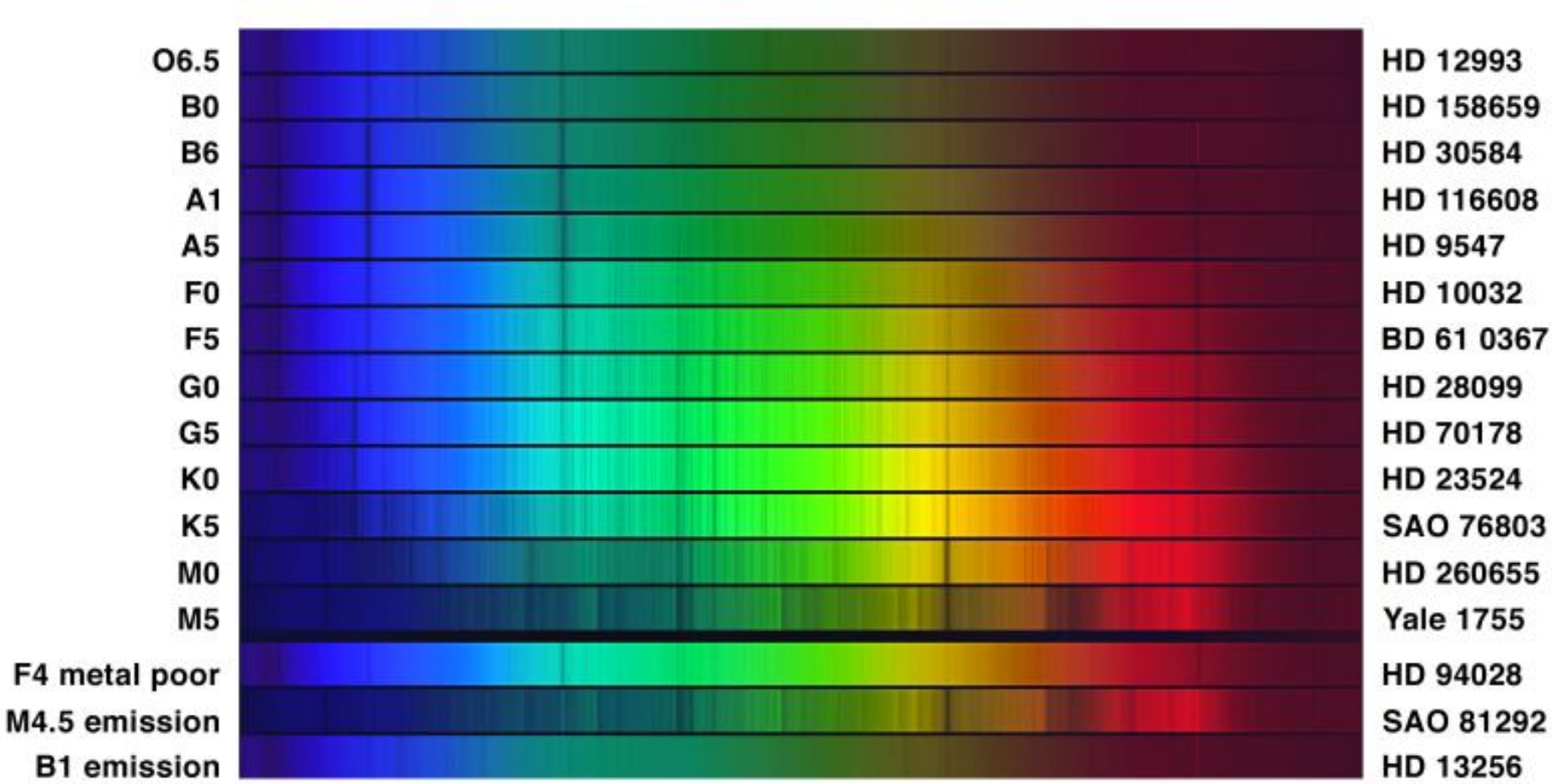
Source: Wikimedia Commons



Stellar spectra across MK classification

Credit: NASA

Spectral type, peculiarities and abundances



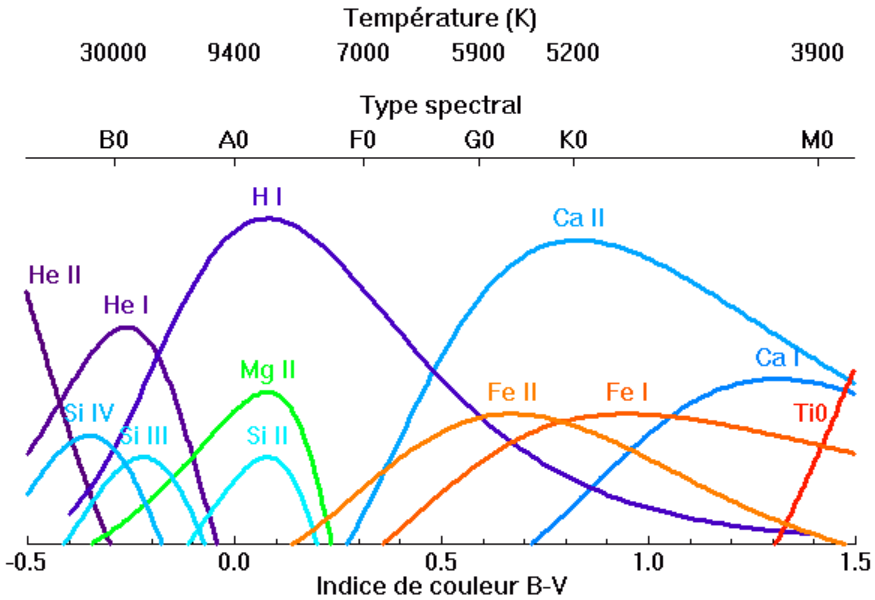
Stellar spectra across MK classification

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Spectral type, peculiarities and abundances

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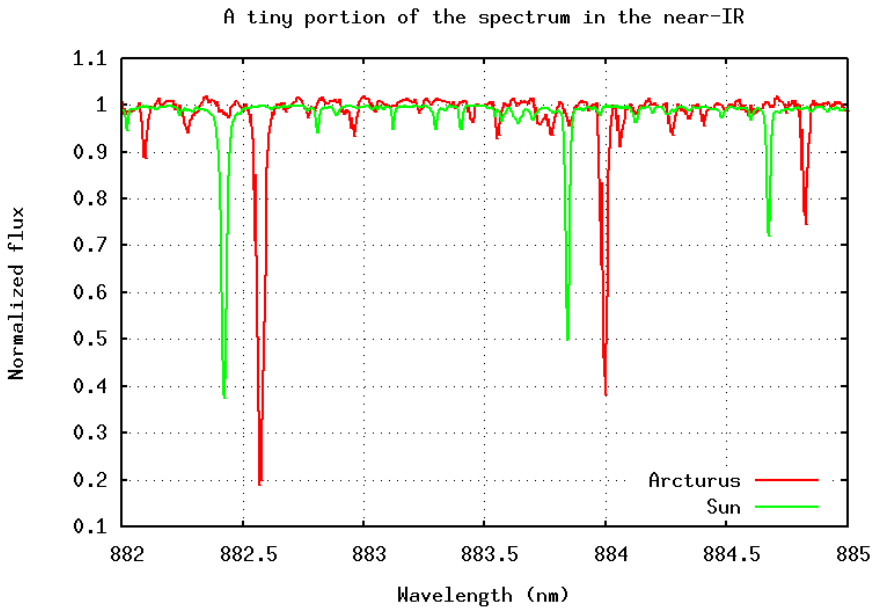
➔ p, m, e, . . .



Credit: «Astrophysique sur Mesure»
Obs. de Paris / UFE

Radial velocities and redshifts

- Doppler-Fizeau effect
  - $z_D = \dfrac{\lambda_{obs} - \lambda_0}{\lambda_0} = \sqrt{\dfrac{c+v}{c-v}} - 1$
    $\simeq \dfrac{v}{c}$, if $v \ll c$
  - Applications:
    - Line shifts
      proper motion, companion
    - Line broadening
      temperature, convection, rotation
- Gravitational redshift (GR)
- Cosmological redshift
  - $z_c = \dfrac{\lambda_{obs} - \lambda_0}{\lambda_0} = \dfrac{a(t_0)}{a(t_{LE})} - 1$

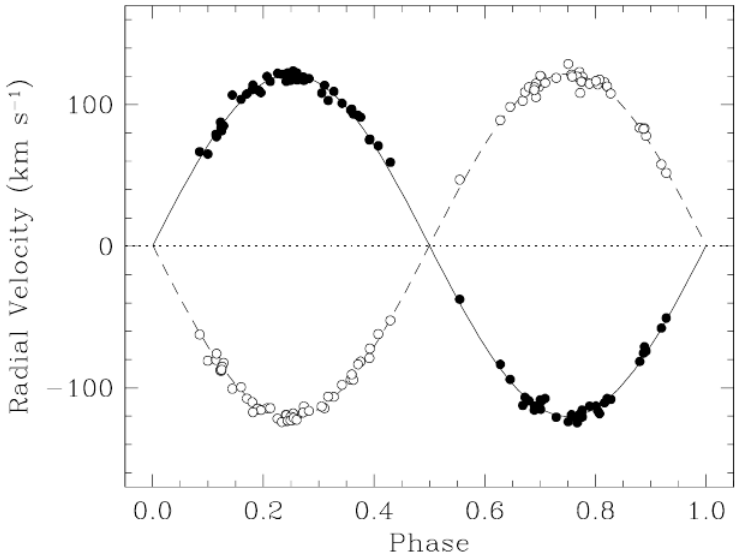


nIR spectra of the Sun and Arcturus

Credit: M. Richmond

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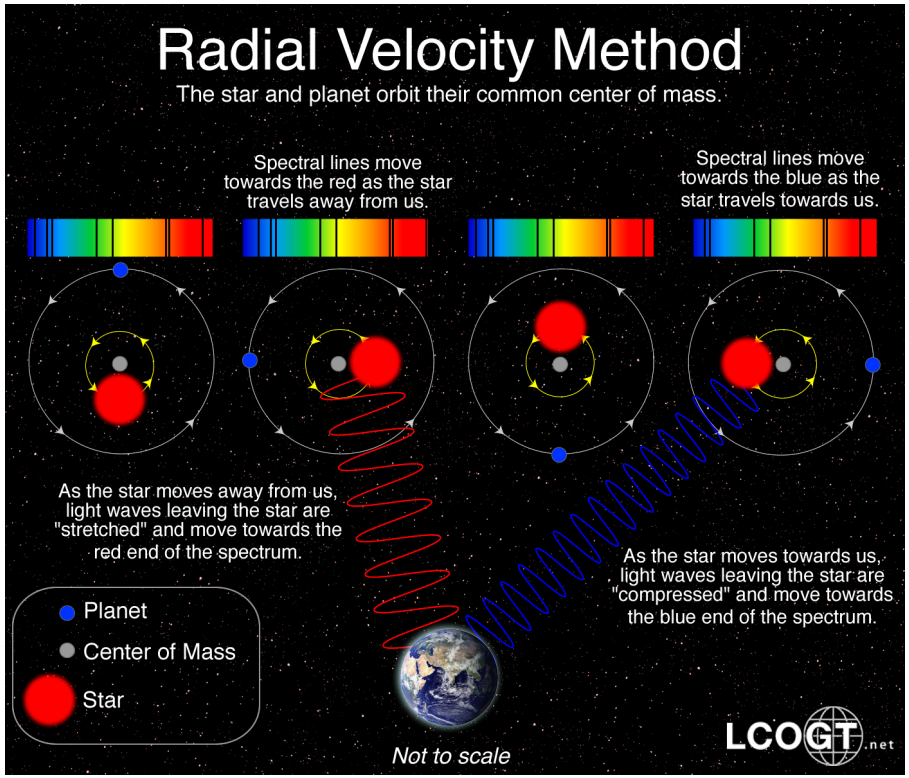
Element	Value
Adjusted quantities:	
P (days)[a]	0.814282212
γ (km s^{-1})	0.54 ± 0.26
K_A (km s^{-1})	121.18 ± 0.42
K_B (km s^{-1})	120.51 ± 0.42
e[b]	0
Min I (HJD −2,400,000)[a]	49,345.112327
Derived quantities:	
$M_A \sin^3 i$ ($M_\odot$)	0.5938 ± 0.0046
$M_B \sin^3 i$ ($M_\odot$)	0.5971 ± 0.0046
$q \equiv M_B/M_A$	1.0056 ± 0.0050
$a_A \sin i$ (×10^6 km)	1.3569 ± 0.0047
$a_B \sin i$ (×10^6 km)	1.3493 ± 0.0047
$a \sin i$ ($R_\odot$)	3.8882 ± 0.0095
Other quantities pertaining to the fit:	
N_{obs}	90
Time span (days)	456.9
σ_A (km s^{-1})	3.44
σ_B (km s^{-1})	3.44

Radial velocities of YY Gem

Torres & Ribas (2002)

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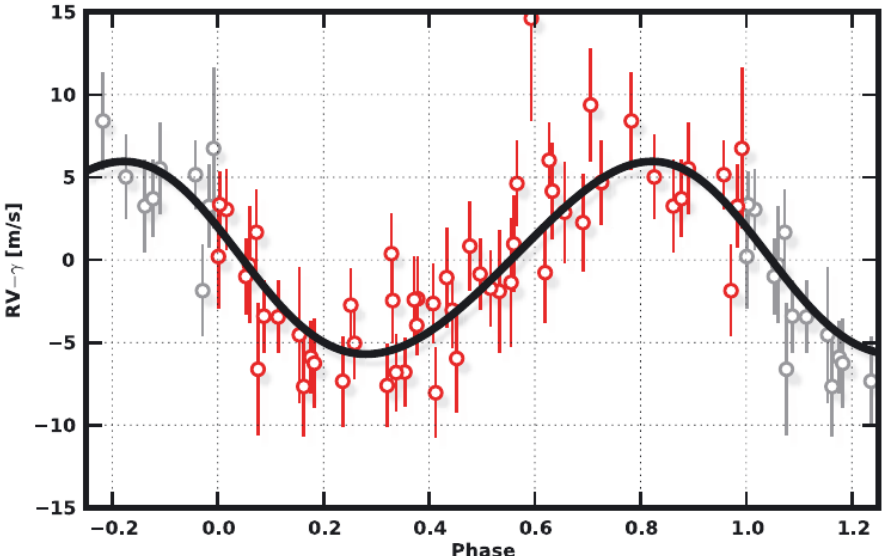


Detecting extrasolar planets

Credit: Las Cumbres Observatory Global Telescope Network

Radial velocities and redshifts

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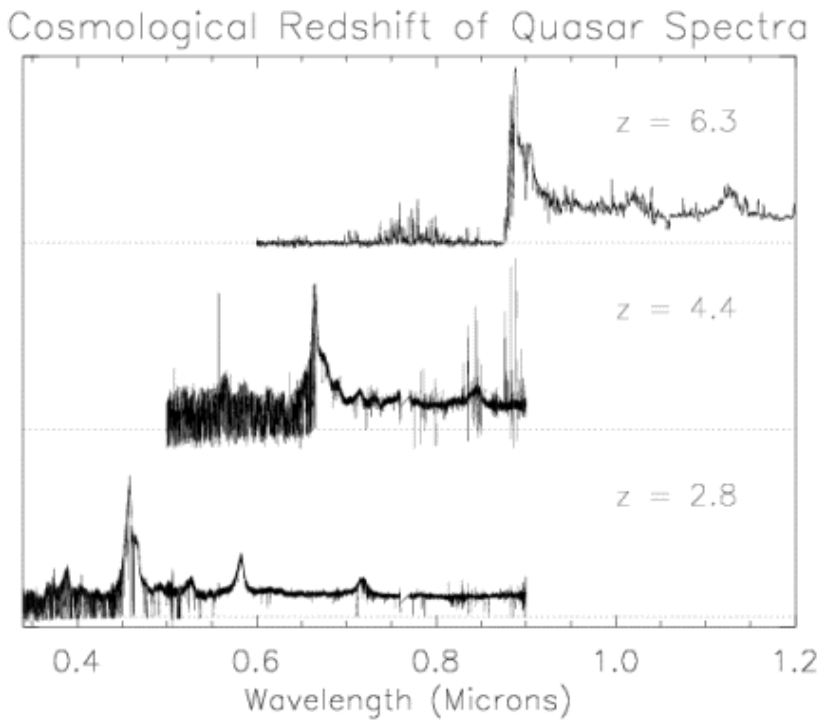
Deduced parameters		
RV K	5.59 ± 0.55	m/s
Orbital semi-major axis a	$0.0287^{+0.0010}_{-0.0011}$	AU
Orbital inclination i	59^{+18}_{-24}	deg
Orbital eccentricity e	$0.080^{+0.094}_{-0.057}$	
Upper limit to e (99%)	<0.36	
Argument of periastron ω	84^{+52}_{-130}	deg
$M_p \sin i$	$7.0^{+0.9}_{-0.8}$	$M_\oplus$
M_p	$8.4^{+4.0}_{-1.5}$	$M_\oplus$

Detection of GJ 3634b

Bonfils et al. (2011)

Radial velocities and redshifts

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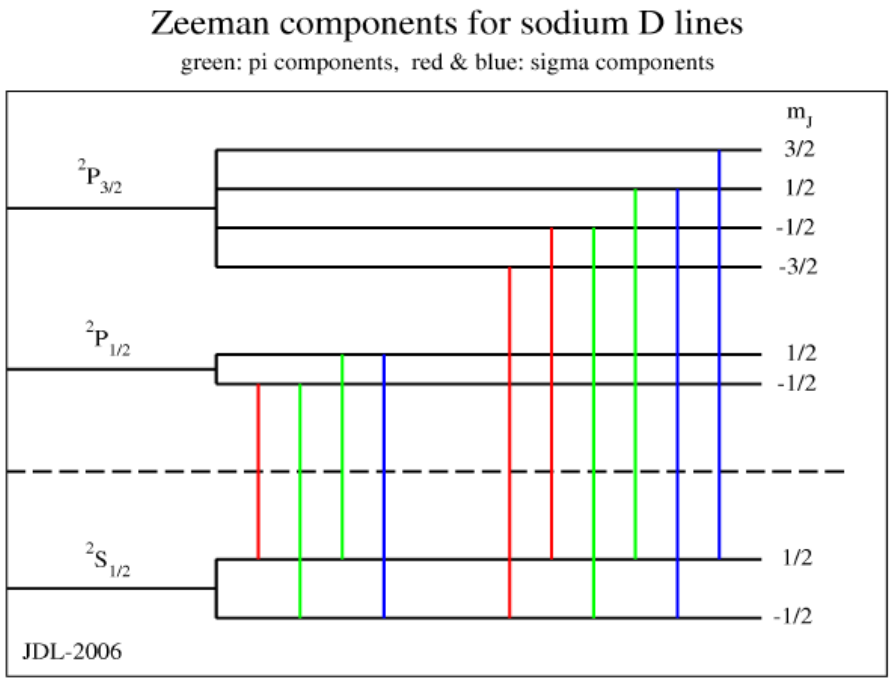


Cosmological redshifts of quasars

Credit: R. Simcoe

Magnetic field measurements

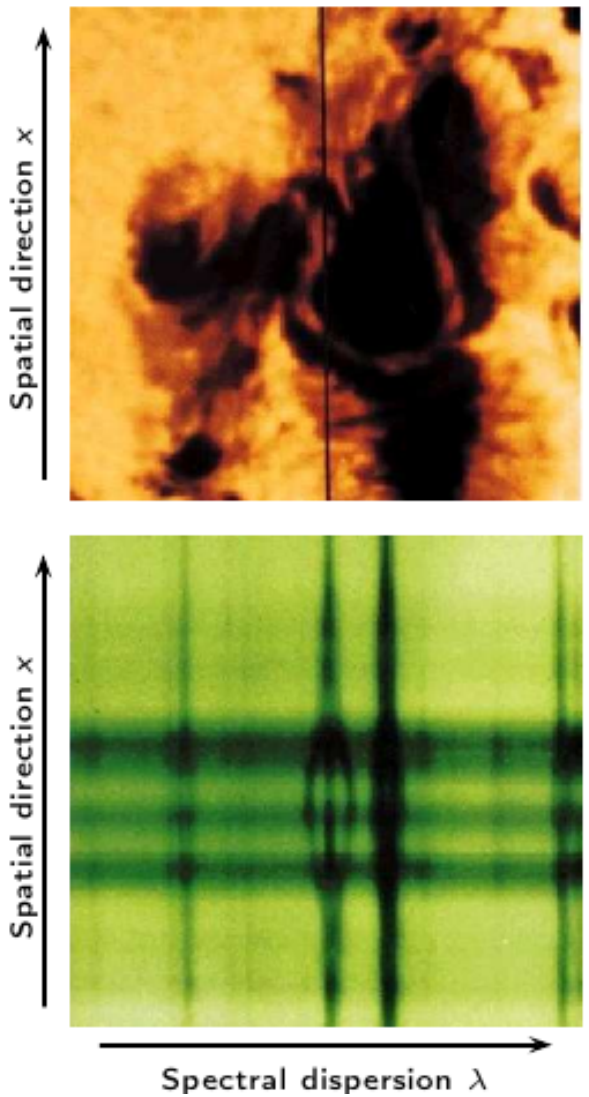
- Zeeman effect
  - **B** breaks J-degeneracy
  - Selection rules
    - 3 component σ_b, π, σ_r
  - Zeeman π-to-σ splitting
    - $\Delta\lambda_B = \dfrac{e}{4\pi m_e c}\lambda_0^2 g_{eff} B$
      $= 4,67\times10^{-12}\,\lambda_0^2 g_{eff} B$
      $B(\mathrm{G})$; $\lambda(\mathrm{nm})$
  - Polarization
    - Information on vector properties of **B**
- Applications
  - Optical, IR
    - Stellar **B**
  - Radio
    - circumstellar env., ISM



Credit: J. Landstreet

Magnetic field measurements

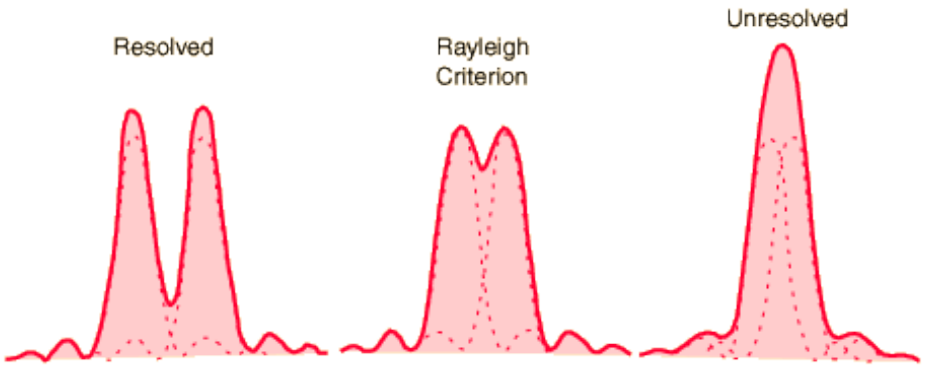
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Credit: NOAO

Importance of spectral resolution

- Spectral resolution
  - Ability to distinguish neighbouring spectral features
    - Analogous to angular resolution
  - Velocity resolution
    - $R = \frac{\lambda}{\Delta\lambda} \Longrightarrow \Delta v = \frac{c}{R}$
- Effect of finite R
  - Convolution with instrumental profile
- Resolved spectral feature
  - width $\sim$ intrinsic

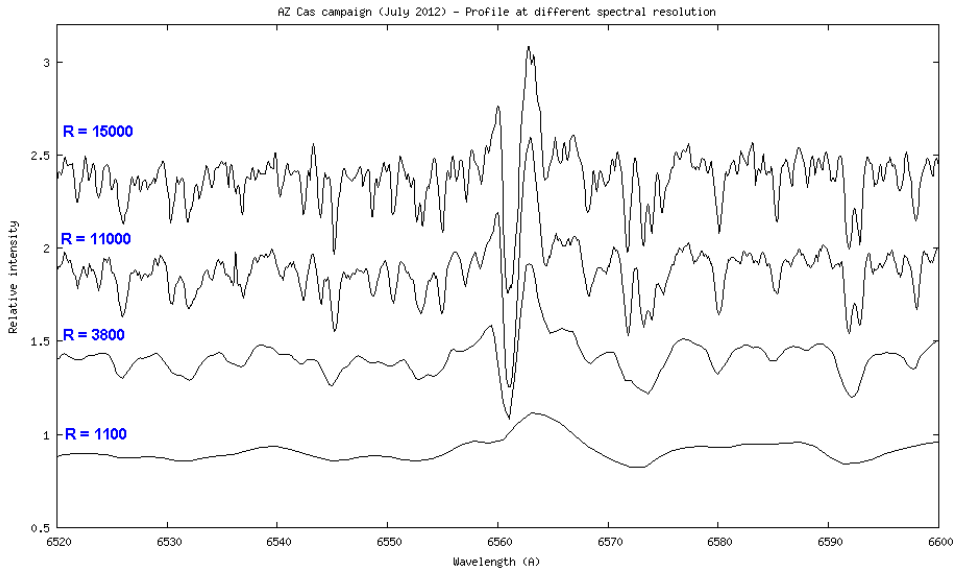


Angular resolution

Credit: Hyperphysics

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Effect of spectral resolution

Credit: Ch. Buil

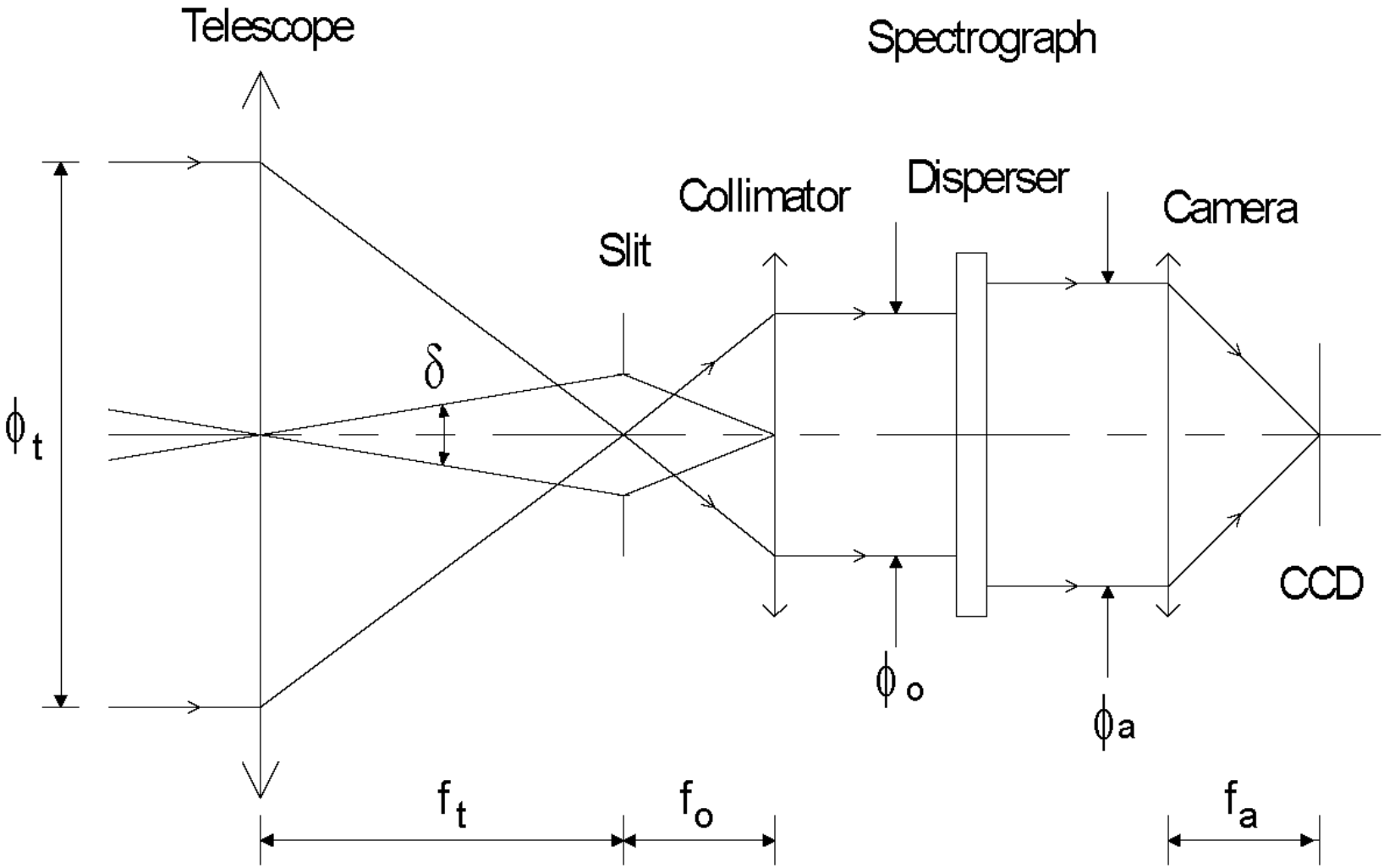
Outline

1 Spectroscopy in Astrophysics

2 Optical-range spectrometers
- Basic layout
- Main parameters
- Dispersive element
- Other spectrograph designs

3 Bibliographic ressources

Basic layout

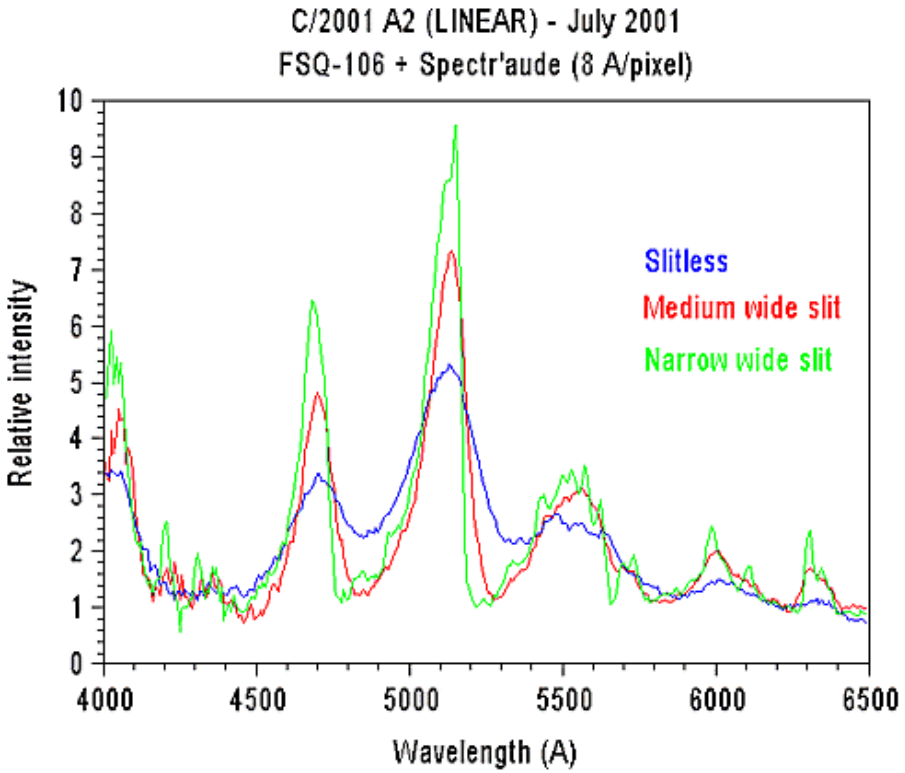


Basic layout of a slit spectrograph

Credit: CAOS group

Main parameters

- Disperser
  - Sets resolution $R = \dfrac{\lambda}{\Delta\lambda}$
  - Sets angular dispersion $\dfrac{dj}{d\lambda}$
- Slit width
  - Adapted to telescope diameter
    - Avoid R degradation
  - Flux – R trade-off (non-DL case)
- Detector
  - Size match λ domain
  - Sampling match R
    - Nyquist criterion

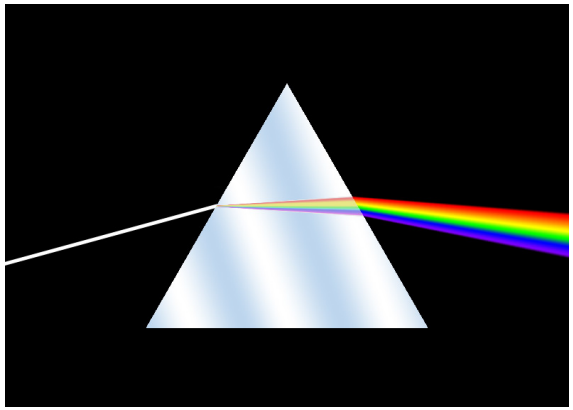


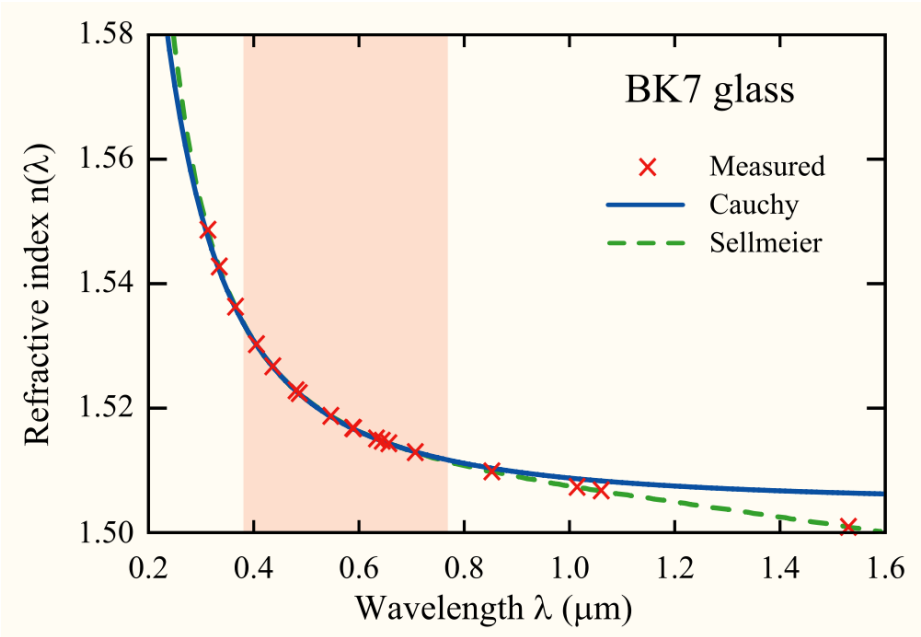
Effect of slit width

Credit: C. Buil

Dispersive element

- Prism
  - $n(\lambda) \Rightarrow$ dispersion
  - $R = b\dfrac{dn}{d\lambda}$
- Diffraction grating
  - N-slits interferences
  - transmission, reflexion
  - blazed grating
  - $R = mN < \dfrac{G}{\lambda}$
    - m: order
    - N: # of grooves
    - G: width
- Other dispersers
  - Volume-phase grating
  - Echelon grating
  - Grism



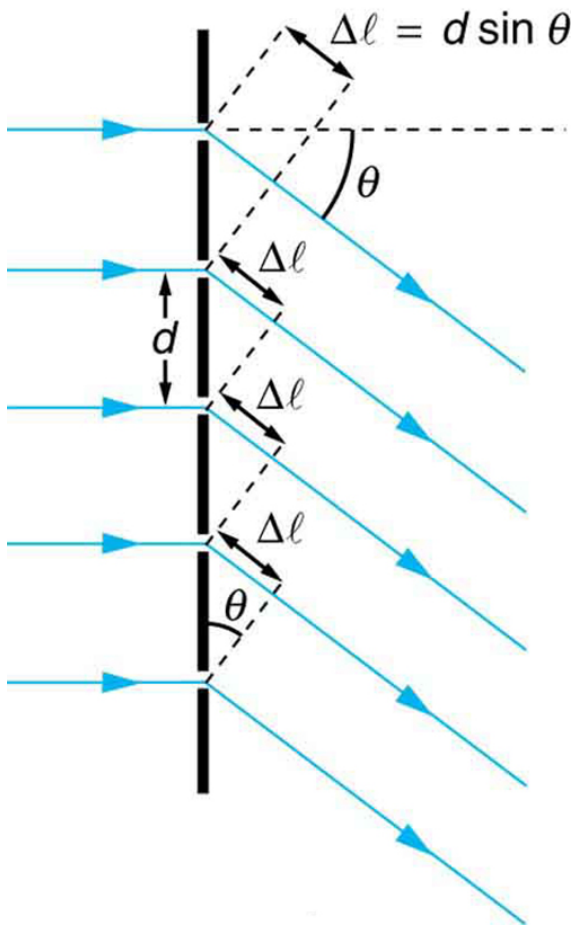


Refractive index of BK7

Source: Wikimedia

Dispersive element

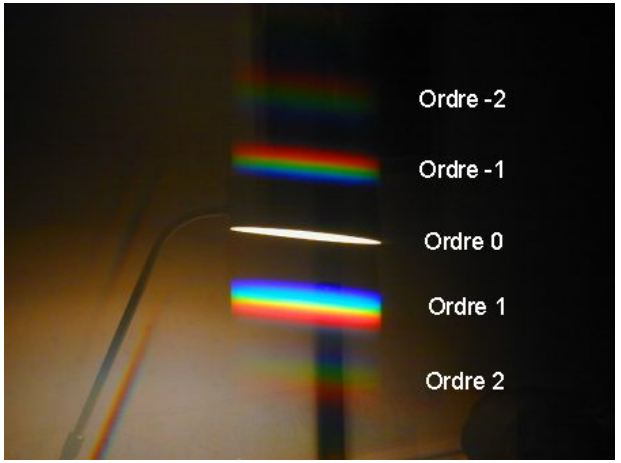
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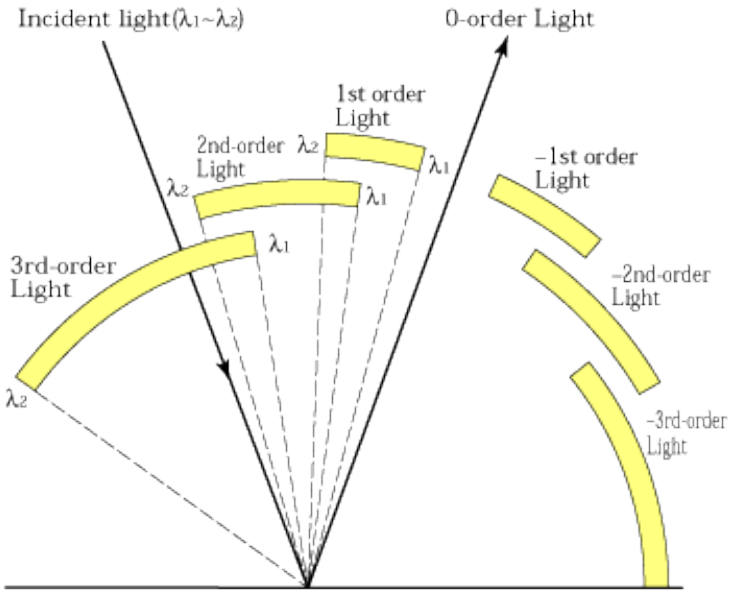
Grating: optical path differences
Source: OpenStax College

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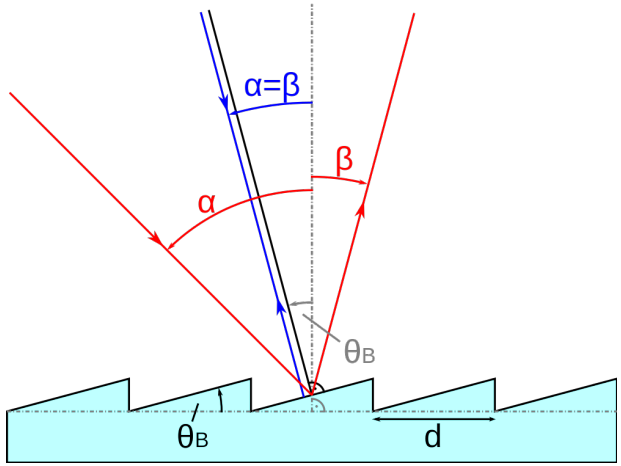
Credit: C. Buil



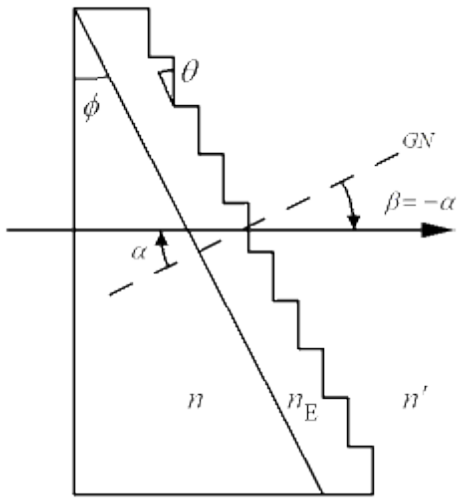
Grating order overlap
Source: Shimadzu

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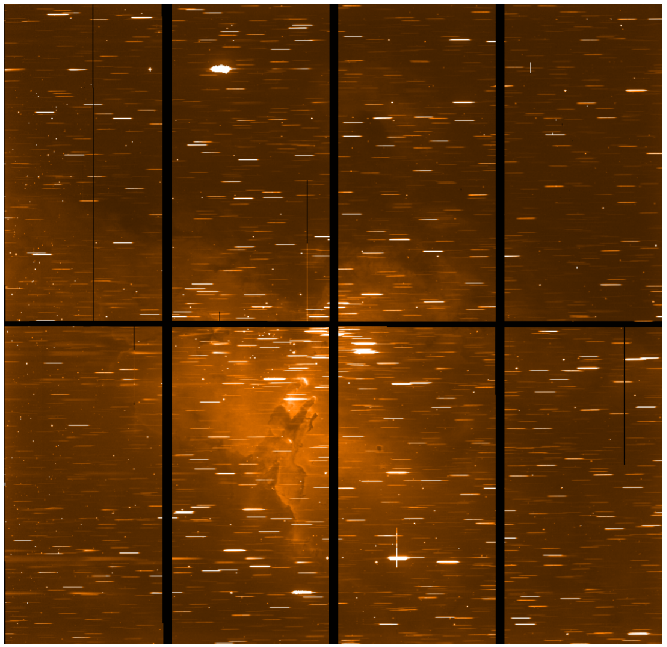
Blazed grating
Source: Wikimedia



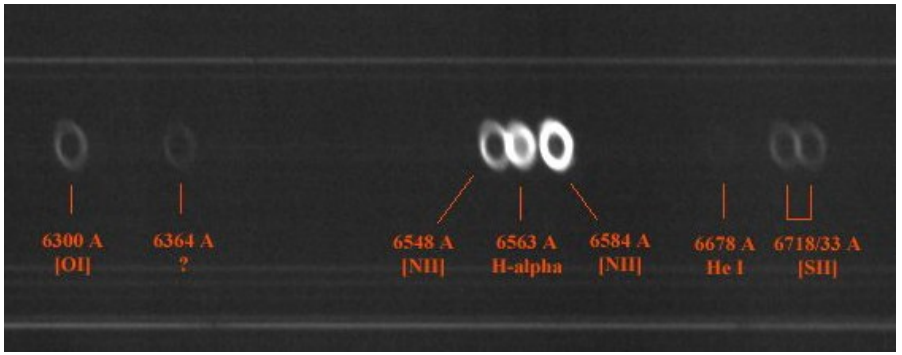
Grism
Source: Richardson Gratings

Other spectrograph designs

- Slitless spectrograph
  - Pros: Multi-object
  - Cons:
    Limited R, overlap, point sources
- Echelle cross-dispersed spectrograph
  - High-R w/o order overlap
  - Wide λ-coverage
- Multi-object spectrograph
  - Fiber coupled
  - Integral field unit
- Other techniques
  - Based on Fabry-Perot
  - Fourier transform spectrograph
  - Detection of photon energy



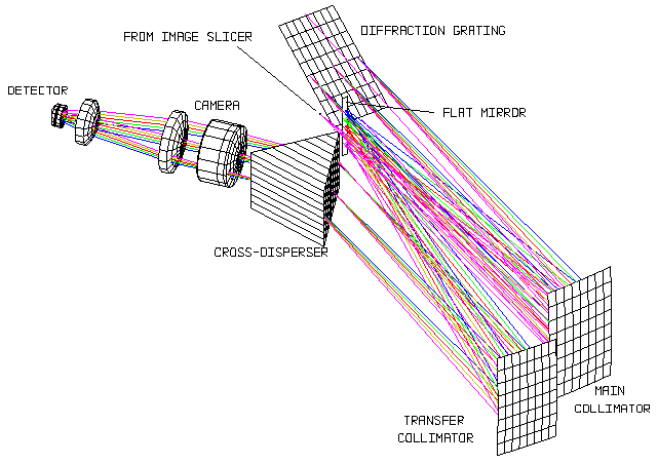
M16 slitless spectroscopy with ESO-WFI
Martayan et al. (2008)



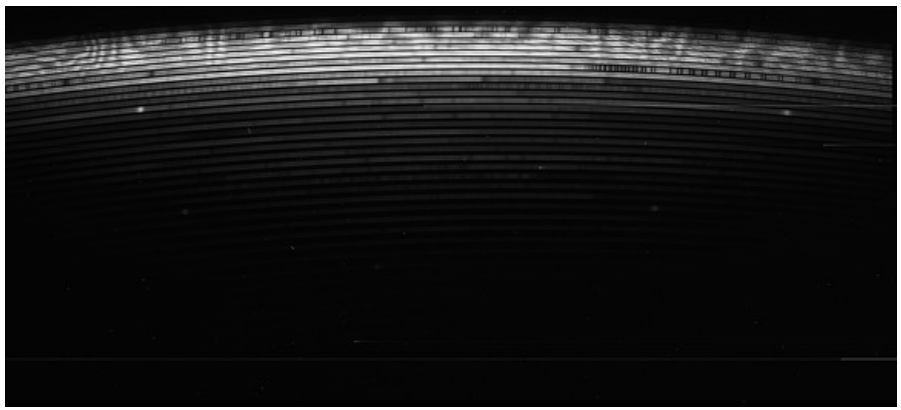
Slitless spectrum of M57
Credit: V. Dhillon

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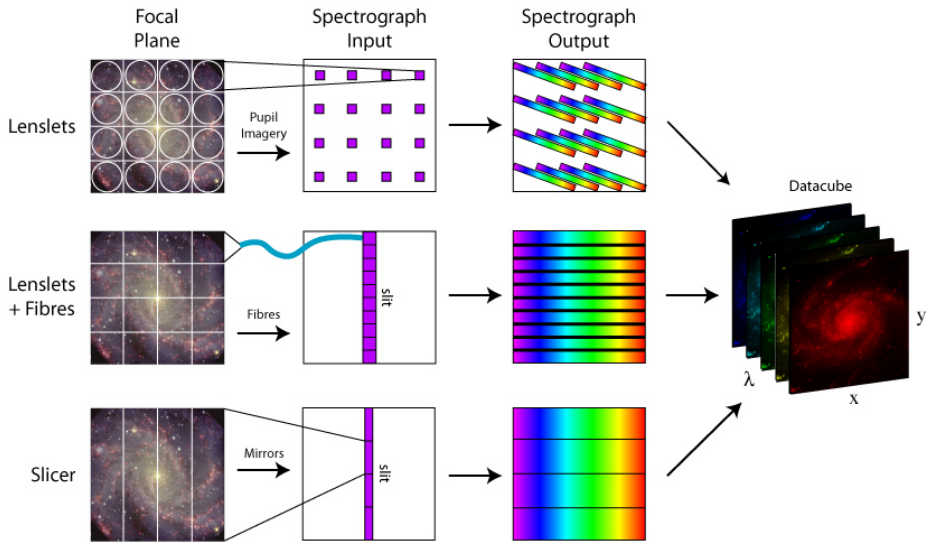
Optical layout of CFHT/ESPaDOnS
Credit: J.-F. Donati



ESPaDOnS spectrum of an M dwarf

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Integral Field spectroscopy
M. Westmoquette, adapted from
Allington-Smith et al. (1998)

Outline

1 Spectroscopy in Astrophysics

2 Optical-range spectrometers

3 Bibliographic ressources

Bibliographic ressources

- *"Astronomie Astrophysique, Introduction, 5e édition"*, A. Acker, 2013, Dunod
  ISBN-13: 978-2100576401
  - Chapitre 2 : "Déchiffrer le rayonnement des astres"
  - Chapitre 4 : "Connaître les étoiles"
- *"Introduction to Astronomical Spectroscopy"*, I. Appenzeller,
  Cambridge Observing Handbooks for Research Astronomers (Book 9),
  Cambridge University Press, 2013
  ISBN-13: 978-1107601796
- *"L'Observation en Astrophysique"*, P. Léna et al., 2008, EDP Sciences
  ISBN-13: 978-2868838773
  - Chapter 8 : "L'analyse spectrale"
- *"Spectroscopie atomique – Instrumentation et structures atomiques"*,
  É. Biémont, 2006, de Boeck
  ISBN-13: 9782804150358
- *"Planets, Stars and Stellar Systems - Volume 2: Astronomical Techniques, Software, and Data"*, Eds T.D. Oswalt & H.E. Bond, 2013
  - Chapitre 2 : "Astronomical Spectroscopy"